

TRAVEC: TRANSITION AND RECOVERY ALIGNMENT VIA VERIFIABLE EVIDENCE-GROUNDED CONTRACTS FOR MOBILE GUI AGENTS

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ABSTRACT

Recent advances in vision-language models are turning GUI agents from interface perception and element grounding systems into general agents capable of autonomously executing real-world user tasks. However, long-horizon interaction remains vulnerable to silent action failures, state drift, repetitive behaviors, and inadequate recovery, while frontier agents often mitigate these problems with lengthy reasoning, iterative reflection, or multi-model verification at every step, creating inference latency and computational costs that accumulate with task length and have become a major barrier to practical deployment. We introduce TRAVEC, a framework for transition and recovery alignment via verifiable evidence-grounded contracts, which systematically embeds state abstraction, transition prediction, outcome verification, and failure diagnosis into data construction and reinforcement learning so that the deployed Action Policy only needs to emit executable actions. TRAVEC provides an automated annotation and generation pipeline that unifies heterogeneous GUI trajectories and combines deterministic state differencing, evidence-grounded multimodal annotation, and counterfactual recovery generation to produce transition-contract, negative-recovery, preference, and verifier-calibration supervision at scale. Building on these data, TRAVEC trains two auxiliary adapters, a Contract Policy and a Verifier, uses them to create auditable self-generated experience through resettable predict–execute–verify–recover rollouts, and optimizes the Action Policy with offline GRPO using six rewards: Action Type, Action Argument, and our four proposed rewards for State Delta, Recovery, Progress/Loop, and Online Environment. Starting from Qwen3.5-9B, we train TRAVEC-9B and compare it with seven frontier models on six benchmarks, with extensive ablations further validating its components; the results demonstrate clear advantages in both task performance and inference efficiency.

1 INTRODUCTION

Graphical user interfaces (GUIs) are the primary gateway through which people access mobile applications, desktop software, and web services, and therefore provide a natural substrate for general agents to turn language intentions into consequential digital actions (Wang et al., 2025b; Tang et al., 2025). Foundation-model-based GUI agents combine interface perception, language understanding, task planning, and action generation to execute multi-step instructions using generic operations such as clicking, swiping, and typing (Wang et al., 2025b; Tang et al., 2025). CogAgent and SeeClick showed that directly modeling small interface elements and spatial locations can reduce reliance on HTML, accessibility trees, or application-specific APIs (Hong et al., 2024; Cheng et al., 2024). OS-World and AndroidWorld subsequently moved evaluation from static element localization to reproducible interaction with real operating systems, where cross-application behavior, long trajectories, and execution outcomes can be checked by the environment (Xie et al., 2024; Rawles et al., 2025). Recent foundation GUI agents further share environments, data, and action representations across

mobile, desktop, and web platforms, marking a shift from local GUI grounding toward general digital task execution (Tang et al., 2025; Zhou et al., 2026a).

This capability has rapidly progressed from perception, to decision making, and then to environment-aligned learning. In 2024, CogAgent and SeeClick improved interface understanding and action localization with high-resolution visual encoding and large-scale GUI grounding data (Hong et al., 2024; Cheng et al., 2024). Mobile-Agent-v2 and Agent S then organized task decomposition, progress memory, experience retrieval, and post-action reflection into multi-role or hierarchical online decision processes for longer horizons (Wang et al., 2024; Agashe et al., 2025a). In 2025, Aguis and UI-TARS incorporated cross-platform actions, planning, reflection, and System-2 reasoning into native visual GUI models, supported by large trajectory collections and reflective online data (Xu et al., 2025b; Qin et al., 2025). In 2026, ComputerRL scaled end-to-end online reinforcement learning over parallel desktop environments, while MobileForge connected real-app exploration, automatic task generation, hierarchical feedback, and step-level GRPO in an annotation-free adaptation loop (Lai et al., 2026; Liu et al., 2026b). Two coupled challenges nevertheless remain. The first is *long-horizon reliability*: a syntactically plausible action or coordinate does not establish that the interface changed as intended, and silent failures, state drift, and accumulated errors remain difficult to audit and recover from (Rawles et al., 2025; Yu et al., 2026). The second is *inference efficiency*, now a primary obstacle to practical GUI-agent deployment. To obtain higher success rates, frontier systems often introduce longer step-wise reasoning and additional reflection during execution (Qin et al., 2025; Huang et al., 2026).

Our central idea is to turn state abstraction, consequence prediction, outcome verification, and failure diagnosis—computations usually performed through online chain-of-thought, self-reflection, or external checking—into scalable data supervision and reinforcement-learning signals. We introduce TRAVEC and use a *transition contract* as the intermediate representation of this process. Each contract binds a task-relevant state, action preconditions, a normalized action, the expected state change, observable evidence, and a recovery branch. We first map AndroidControl, AITW, AMEX, CMGUI, MobileViews, and grounding datasets into a unified transition schema, then extract deterministic signals for page identity, focus, input slots, blocking events, text changes, and no-change outcomes. An evidence-grounded multimodal annotator constructs contracts, hard negatives, counterfactual recoveries, preferences, and verifier-calibration examples; JSON-schema, action-equivalence, coordinate-bound, and evidence-grounding checks enforce quality. Crucially, the pipeline separates information available at decision time from post-action evidence: the next observation is used to generate and verify labels but never enters the Action Policy input. This produces 550K item-level, leakage-free, verifiable annotations.

Starting from Qwen3.5-9B, TRAVEC trains an Action Policy that emits exactly one normalized action and independently branches two training-time adapters: a Contract Policy that predicts a transition before execution and a Verifier that evaluates the observed outcome afterward. In resettable Android environments, each task is executed from the same snapshot in four rollouts. The Contract Policy and Verifier provide pre- and post-action checks, while a trajectory Critic assesses task outcome, step reasonableness, and non-oracle recovery hints, forming a predict–execute–verify–recover audit loop. Reliable self-generated steps are used for offline GRPO with Action Type and Action Argument rewards adapted from MobileForge (Liu et al., 2026b), together with four rewards proposed here: State Delta, Recovery, Progress/Loop, and Online Environment. At deployment, the Contract Policy, Verifier, and Critic are removed; TRAVEC-9B directly produces action-only outputs. We evaluate the resulting model on six static and dynamic benchmarks against seven frontier GUI agents and conduct component, data, and reward ablations.

Our contributions are threefold:

- We develop a scalable automated GUI annotation and generation pipeline. Its transition contracts jointly represent task state, preconditions, expected changes, observable evidence, and failure recovery, enabling 550K item-level verifiable annotations through deterministic differencing, evidence-grounded labeling, and counterfactual generation.
- We introduce a training-time process-audit and recovery mechanism built from Contract and Verifier adapters plus a trajectory Critic. It extracts leakage-free offline GRPO data from self-generated interaction and aligns the policy with six rewards covering action protocol, state transition, recovery, task progress, loop avoidance, and real execution outcomes.

- We distill structured reasoning, verification, and recovery into TRAVEC-9B while retaining a single action-only policy at deployment. Evaluation on six benchmarks, seven frontier models, and systematic ablations examines both task performance and inference efficiency.

2 RELATED WORK

2.1 GUI GROUNDING AND NATIVE ACTION MODELS

Reliable GUI control begins with mapping a linguistic intent to a small, visually ambiguous interface target. CogAgent introduced high-resolution cross-module visual processing (Hong et al., 2024); SeeClick paired automatically curated grounding data with ScreenSpot (Cheng et al., 2024); and Ferret-UI specialized region-aware understanding for elongated mobile screens (You et al., 2024). OmniParser instead converts screenshots into interactable, semantically labeled regions (Lu et al., 2024). UGround and OS-Atlas scaled visual grounding to millions of cross-platform screenshots and elements (Gou et al., 2025; Wu et al., 2025), while ShowUI reduced redundant visual tokens in a lightweight vision-language-action model (Lin et al., 2024). Aguis, UI-TARS, ScaleCUA, and the GUI-Owl models of Mobile-Agent-v3/v3.5 integrated grounding, planning, and unified action modeling into native agents trained on broader trajectory mixtures (Xu et al., 2025b; Qin et al., 2025; Liu et al., 2026c; Ye et al., 2025; Xu et al., 2026). These models strengthen action prediction, but correct action form and localization do not verify that the environment realized the intended transition. TRAVEC complements action modeling with evidence-grounded transition and recovery supervision while preserving an action-only interface.

2.2 LONG-HORIZON PLANNING, MEMORY, AND RECOVERY

Long-horizon systems commonly add structure at inference time. Mobile-Agent-v2 separates planning, decision, and reflection agents (Wang et al., 2024); AppAgent v2 builds a retrievable interaction knowledge base (Li et al., 2025b); and Agent S/S2 combine hierarchical planning, experience retrieval, and specialists (Agashe et al., 2025a;b). UI-TARS and UI-TARS-2 train reflective or multi-turn reasoning traces (Qin et al., 2025; Wang et al., 2025a). HiconAgent learns history compression (Zhou et al., 2026b), whereas MemGUI-Agent, UI-Mem, and UI-Copilot maintain structured experience or invoke auxiliary retrieval and calculation tools (Liu et al., 2026a; Xiao et al., 2026; Lu et al., 2026). These approaches improve planning and recovery, but retain history rewriting, reflection, or additional model/tool calls on the execution path. TRAVEC instead uses recovery as a supervised contract field and an RL process signal, and removes auxiliary reasoning modules at deployment.

2.3 AUTOMATED EXPERIENCE, REINFORCEMENT LEARNING, AND EFFICIENT EXECUTION

Automated data and environment learning have become central to GUI agents. OS-Genesis synthesizes tasks retrospectively from exploration (Sun et al., 2025); GUI-360° builds large LLM-assisted desktop corpora (Mu et al., 2025); and OpenMobile decouples exploration, task synthesis, and policy-switched rollout (Cheng et al., 2026). GUI-R1 and AgentCPM-GUI apply GRPO with action-rule rewards (Luo et al., 2025; Zhang et al., 2025); GUI-Libra addresses partial verifiability (Yang et al., 2026); UI-S1 approximates multi-turn online signals on offline trajectories (Lu et al., 2025b); and ComputerRL and MobileRL scale online environment learning (Lai et al., 2026; Xu et al., 2025a). MobileForge combines generated tasks with hierarchical feedback (Liu et al., 2026b), while OS-Themis uses milestone evidence and multi-agent review to improve GUI outcome rewards (Li et al., 2026). For efficiency, GUI-KV compresses visual history (Huang et al., 2025), MobileExplorer overlaps inference with UI exploration (Huang et al., 2026), and anticipatory policy trees move decoding outside transient decision windows (Dong et al., 2026). In contrast, TRAVEC jointly annotates a predicted transition, observable evidence, and recovery; calibrates a post-action verifier; and converts their audit into offline optimization. Its efficiency comes from moving these computations into training rather than retaining a complex inference-time scaffold.

3 METHOD

TRAVEC places structured transition reasoning and verification in data construction and policy optimization, while deployment retains only the Action Policy. Section 3.1 defines the decision problem and transition contract; Section 3.2 presents the automated data pipeline; Section 3.3 trains three responsibility-separated adapters; and Section 3.4 describes process-audited offline GRPO.

3.1 PROBLEM FORMULATION

We model goal-conditioned GUI interaction as a partially observable decision process $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathcal{A}, P, \Omega, \mathcal{G}, \mathcal{Y})$. At step t , the environment record $o_t = (I_t, U_t, X_t)$ may contain a screenshot, a UI hierarchy, and OCR or other structured evidence for data construction and auxiliary verification. The deployed Action Policy receives only the screenshot I_t . Given goal g , action history $h_t = (a_1, \dots, a_{t-1})$, and an optional hint η_t derived only from an earlier failed attempt, it predicts

$$\pi_{\theta_A}(a_t \mid g, I_t, h_t, \eta_t), \quad a_t = (\alpha_t, \psi_t), \quad (1)$$

where α_t is an action type and ψ_t its type-specific argument. The normalized type set includes click, long press, swipe, type, key/system button, open, wait, and terminate. A trajectory $\tau = (g, o_1, a_1, \dots, o_T, a_T, o_{T+1})$ receives an execution-based outcome $Y(\tau) \in \{0, 1\}$. We seek $\theta_A^* = \arg \max_{\theta_A} \mathbb{E}[Y(\tau)]$ subject to the deployment constraint that each step emits exactly one parseable action and invokes no auxiliary model.

Let $d_t = \mathcal{D}(o_t, o_{t+1})$ be a deterministic observable state difference containing page/activity, text, focus, input, task-slot, blocking-event, and no-change fields. Rather than predict future pixels, we describe the task-relevant local consequence with a transition contract

$$c_t = (\tilde{s}_t, p_t, a_t, \hat{d}_t, E_t, \rho_t), \quad \rho_t = (\phi_t, a_t^{\text{rec}}, \hat{d}_t^{\text{rec}}, E_t^{\text{rec}}). \quad (2)$$

Here $\tilde{s}_t$ is a compact task state, p_t the action precondition, $\hat{d}_t$ the expected difference, E_t up to three checkable evidence queries, and ρ_t an optional recovery trigger, action, expected difference, and evidence. During training, a Contract Policy predicts $q_{\theta_C}(c_t \mid g, o_t, h_t, a_t)$ before execution, while a Verifier predicts $V_{\theta_V}(y_t^V \mid g, o_t, c_t, o_{t+1}, \hat{d}_t, \chi_t)$ afterward. The deterministic checks χ_t cover parsing, coordinates, enumerations, and action consistency. The verifier output $y_t^V = (\mathbf{v}_t, \nu_t, \kappa_t)$ contains eight 0–4 scores, an accept/partial/reject verdict, and 0–4 confidence. Neither auxiliary mapping is deployed.

3.2 AUTOMATED DATA ANNOTATION AND GENERATION

Unified transition records. We normalize AndroidControl, AITW, AMEX, CMGUI, MobileViews, ScreenSpot, and ScreenSpot-Pro into records $x_t = (g, o_t, h_t, a_t, o_{t+1}, y_t^{\text{term}}, m_t^{\text{src}})$. Within this annotation pipeline, ScreenSpot and ScreenSpot-Pro are assigned exclusively to the validation and test splits of the Contract Policy and Transition Verifier and are excluded from all training splits. Coordinates are stored in $[0, 1]^2$ and deterministically rendered to integer coordinates in $[0, 1000]^2$; swipes follow touch-to-lift semantics. These static grounding samples assess localization but are not treated as strong transition evidence. A deterministic extractor first computes a weak current state $\bar{s}_t = F_{\text{state}}(g, o_t)$ and, when a next observation exists, the structured difference $d_t = \mathcal{D}(o_t, o_{t+1})$. Perceptual hashes, OCR, UI structure, page/activity identity, focus, editable fields, keyboard or permission overlays, and task slots jointly support machine-checkable evidence.

The formal pipeline uses Claude Opus 4.8 as the multimodal annotator, but acceptance is controlled by model-independent validators. The target recipe contains 350K contract/state items, 120K negative-recovery items, 60K preference items, and 20K verifier-calibration items, totaling 550K annotations. Before the full run, a stratified 100-item quality gate requires at least 98% process success, 99% schema success, at most 3% evidence error, 2% action-equivalent negatives, 5% invalid recoveries, and at least 95% independent audit pass rate.

Transition-contract annotation. For a demonstrated transition with adjacent observations, an evidence-grounded annotator produces $c_t^+ = A_C(x_t, \bar{s}_t, d_t)$ without rewriting the demonstrated action. An item is retained only if the contract action is semantically equivalent to a_t and the deterministic validator accepts its wrapper, JSON schema, fields, coordinates, evidence cardinality, and

recovery structure:

$$\text{Eq}_A(a(c_t^+), a_t) = 1, \quad V_{\text{det}}(c_t^+, x_t) = 1. \quad (3)$$

Current-observation evidence is used only for $\tilde{s}_t$ and p_t ; o_{t+1} and d_t supervise the expected difference and evidence. Thus the next frame never becomes policy input.

Counterfactual negative and recovery generation. For each accepted contract, deterministic mutations construct up to four hard-negative actions by perturbing the target, action type, argument, or order while preserving syntactic plausibility. The annotator predicts the likely failed transition, an observable failure trigger, and a minimal recovery branch. We reject a negative if it is action-equivalent to the demonstrated action, unsupported by the current interface, or paired with a recovery that merely repeats the failed action. Real failed transitions, when available, are retained separately from counterfactual failures so that evidence provenance remains explicit.

Preference and verifier calibration. Preference pairs hold the decision input fixed and contrast an accepted contract or recovery with a hard negative. Calibration items combine positive transitions, counterfactual negatives, real failures, and deterministic corruption cases. They supervise eight verifier dimensions: output format, action validity, state grounding, delta evidence, task progress, recovery applicability, loop safety, and cost efficiency. This design makes the verifier sensitive to realized consequences rather than lexical similarity alone.

3.3 SUPERVISED TRAINING OF ROLE-SPECIALIZED ADAPTERS

We parameterize each role as $f_{W_0, \phi}$, a Qwen3.5-9B backbone W_0 augmented by one set of LoRA parameters ϕ . Action SFT first learns ϕ_A^{SFT} , defining the action-only policy $\pi_{\theta_A^{\text{SFT}}} = f_{W_0, \phi_A^{\text{SFT}}}$. The Contract and Verifier branches are then initialized from independent copies of ϕ_A^{SFT} :

$$\phi_C^{(0)} = \phi_A^{\text{SFT}}, \quad \phi_V^{(0)} = \phi_A^{\text{SFT}}. \quad (4)$$

They continue supervised optimization under separate role-specific objectives, yielding $q_{\theta_C} = f_{W_0, \phi_C^*}$ and $V_{\theta_V} = f_{W_0, \phi_V^*}$. Thus the auxiliary branches inherit the interface and action prior learned by Action SFT, but have independent parameters and optimization paths after branching.

Action Policy. The Action Policy generates only `<action>{flat JSON}</action>`. A shared renderer/parser enforces one tag, a fixed field set, type-specific arguments, and coordinate validity. Completion-only cross-entropy is applied to 300K action examples for two epochs at learning rate 2×10^{-5} with rank-16 LoRA. The vision encoder is frozen; LoRA targets language and multimodal projector/merger linear layers.

Contract Policy. Starting from $\phi_C^{(0)}$, the Contract Policy continues SFT on expert contracts, real recoveries, and counterfactual recoveries with loss weights (0.70, 0.20, 0.10). It conditions on (g, o_t, h_t, a_t) but cannot access o_{t+1} or d_t , so every predicted contract is available before action execution. The train/dev/test sets contain 287,840/6,080/6,080 samples; each distributed optimizer batch fixes the three task counts at 156/34/34. Training uses two epochs, learning rate 10^{-6} , rank-16 LoRA, and a 256-token completion limit.

Transition Verifier. The Verifier is independently initialized at $\phi_V^{(0)}$ and continues SFT without passing through the Contract branch. It observes the pre-action state, contract, post-action state, deterministic difference, and checks. Positive, hard-negative, and calibration losses are weighted (0.45, 0.45, 0.10). The formal set contains 77,100 positives, 77,100 hard negatives, and 18,504 calibration samples (172,704 total), with fixed per-batch counts 100/100/24. We use two epochs, learning rate 10^{-6} , rank-16 LoRA, and a 128-token completion limit. Contract and Verifier remain separate models during rollout, preserving their pre-execution prediction and post-execution verification roles.

3.4 PROCESS-AUDITED OFFLINE GRPO

Resettable rollout and audit. The executable pool contains 3,249 tasks from 20 applications. After within-app token-Jaccard deduplication at threshold 0.9, we stratify 400 training and 40 development tasks. Each task is attempted $K = 4$ times from the same clean AndroidWorld snapshot.

At step t , the Action Policy proposes a_t from I_t , the Contract Policy predicts c_t before execution, the environment returns o_{t+1} , and the Verifier evaluates the realized transition. After each attempt, a trajectory Critic labels feasibility, final success, and every step’s reasonableness, then provides only an observed-failure/avoid-repeating hint for the next attempt; it is forbidden from revealing the correct next action.

Let $z_{i,t}^{C,(k)}$ denote Critic reasonableness, $z_{i,t}^{E,(k)}$ deterministic execution validity, and let a high-confidence verifier veto be $b_{i,t}^{V,(k)} = \mathbf{1}[\nu_{i,t}^{(k)} = \text{reject} \wedge \kappa_{i,t}^{(k)} \geq 3]$. A step is retained iff

$$z_{i,t}^{(k)} = z_{i,t}^{C,(k)} z_{i,t}^{E,(k)} (1 - b_{i,t}^{V,(k)}). \quad (5)$$

The verifier can remove a Critic-positive step but never turn a Critic-negative step positive. A verifier parse failure does not veto the step.

Offline sample construction. For task i , we retain tasks whose four-attempt success rate is in $[0, 0.9]$, excluding consistently solved tasks. Among attempts, we select lexicographically by task success, retained-step ratio, and shorter length. Only retained self-generated actions are materialized:

$$\mathcal{D}_{\text{GRPO}} = \{(x_{i,t}^{\text{pol}}, a_{i,t}^*, m_{i,t}) : z_{i,t}^{(k_i^*)} = 1\}. \quad (6)$$

The policy input $x_{i,t}^{\text{pol}} = (g_i, I_{i,t}, h_{i,t}, \eta_i^{(<k_i^*)})$ contains decision-time information only. Contracts, post-action observations, verifier/Critic labels, and execution outcomes remain in metadata $m_{i,t}$ for auditing and reward computation.

Six-component reward. A candidate completion is parsed as $\hat{a} = (\hat{\alpha}, \hat{\psi})$ and compared with retained action $a^* = (\alpha^*, \psi^*)$. Parse or format failure sets all rewards to zero. Following the adaptive action scoring used by MobileForge (Liu et al., 2026b), we separate

$$r_{\text{type}} = \mathbf{1}[\hat{\alpha} = \alpha^*], \quad r_{\text{arg}} = r_{\text{type}} S_{\alpha^*}(\hat{\psi}, \psi^*), \quad r_{\text{align}} = 0.2r_{\text{type}} + 0.8r_{\text{arg}}. \quad (7)$$

S_{α^*} uses distance-decayed point similarity for clicks/long presses, exact direction for swipes, token-set F1 for typing, exact or containment matching for app names, and exact type-specific arguments otherwise. We propose four additional rewards from audited metadata:

$$r_{\text{sd}} = r_{\text{align}}(v_{\text{state}} + v_{\text{delta}})/8, \quad r_{\text{rec}} = r_{\text{align}} v_{\text{recovery}}/4, \quad (8)$$

$$r_{\text{pl}} = r_{\text{align}}(v_{\text{progress}} + v_{\text{loop}})/8, \quad r_{\text{env}} = r_{\text{align}}(z_{\text{exec}} + z_{\text{change}})/2. \quad (9)$$

State Delta checks grounded expected change; Recovery checks applicability to the observed failure; Progress/Loop rewards progress while penalizing repetition; and Online Environment uses actual execution validity plus task success or observable state change. The formal reward is

$$R = 0.10r_{\text{type}} + 0.70r_{\text{arg}} + 0.05r_{\text{sd}} + 0.05r_{\text{rec}} + 0.05r_{\text{pl}} + 0.05r_{\text{env}}. \quad (10)$$

Thus action matching contributes 80% of the reward mass and the four proposed process rewards contribute the remaining 20%.

GRPO and deployment. GRPO continues optimization from the Action SFT policy rather than restarting from the backbone. We freeze the SFT policy as the reference and initialize the trainable actor to the same policy:

$$\pi_{\text{ref}} = \pi_{\theta^{\text{SFT}}}, \quad \pi_{\theta(0)} = \pi_{\text{ref}}. \quad (11)$$

The actor is optimized with rank-64 LoRA while the reference remains fixed. For each prompt, $G = 5$ completions receive group-normalized advantages $A_j = (R_j - \mu_R)/(\sigma_R + 10^{-4})$. We optimize the clipped GRPO objective with $\epsilon = 0.2$ and KL coefficient $\beta_{\text{KL}} = 0.01$:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E} \left[\frac{1}{G} \sum_{j=1}^G \min(\xi_j A_j, \text{clip}(\xi_j, 1 - \epsilon, 1 + \epsilon) A_j) \right] + \beta_{\text{KL}} D_{\text{KL}}(\pi_{\theta} \| \pi_{\text{ref}}). \quad (12)$$

Training runs for four epochs at learning rate 10^{-6} on 56 A100 GPUs; an optimizer update contains 112 unique prompts and 560 completions. It is offline because policy optimization reads frozen

Table 1: Offline benchmark performance and AndroidControl inference efficiency. AndroidControl reports success rate (SR). SS and Pro denote target-hit accuracy on ScreenSpot and ScreenSpot-Pro, respectively. Lat. denotes the mean inference time per action (s), and Tok. the mean number of generated tokens per action. Best and second-best results are shown in bold and underlined, respectively.

Model	AndroidControl		GUI-Odyssey			ScreenSpot		Efficiency	
	High $\uparrow$	Low $\uparrow$	TM $\uparrow$	GR $\uparrow$	SR $\uparrow$	SS $\uparrow$	Pro $\uparrow$	Lat. (s) $\downarrow$	Tok. $\downarrow$
Aguvis-7B-720P	57.237	66.914	74.137	70.541	53.373	78.852	20.936	2.853	34.505
GUI-R1-7B	22.464	42.436	70.889	58.397	42.590	<u>85.849</u>	28.716	6.224	32.096
ScaleCUA-7B	36.056	46.608	77.218	69.886	57.315	73.899	23.340	3.591	35.800
GUI-Libra-8B	<u>62.856</u>	<u>81.417</u>	63.181	71.155	43.994	85.535	<u>53.700</u>	5.905	78.294
OpenMobile-8B	52.335	72.502	67.164	59.893	42.332	79.324	38.646	5.763	58.644
MemGUI-8B-SFT	58.061	77.424	68.932	62.140	45.444	76.965	39.216	12.910	318.260
ForgeQwen3-8B	62.665	79.850	79.864	71.221	<u>58.369</u>	82.704	46.490	7.070	86.799
Qwen3.5-9B	34.379	52.966	62.217	<u>90.823</u>	56.185	38.758	42.631	<u>0.815</u>	<u>18.039</u>
TRAVEC-9B (ours)	64.309	93.337	<u>79.588</u>	91.006	72.212	92.689	67.805	0.786	17.152

screenshots and audit metadata rather than re-executing the environment for newly sampled completions. Let θ_A^{TRAVEC} denote the optimized Action Policy. Deployment uses only

$$\pi_{\text{deploy}} \equiv \pi_{\theta_A^{\text{TRAVEC}}}, \quad (13)$$

which emits one normalized action per step. The Contract Policy, Verifier, and Critic are not invoked, and no contract, verifier verdict, recovery analysis, or free-form chain-of-thought is generated.

4 EXPERIMENTS AND RESULTS

4.1 EXPERIMENTAL SETUP

4.1.1 IMPLEMENTATION AND TRAINING CONFIGURATION

We use the post-trained Qwen3.5-9B multimodal model (Qwen Team, 2026) as the common base. Action, Contract, and Verifier SFT use rank-16 LoRA for two epochs; their learning rates are 2×10^{-5} , 10^{-6} , and 10^{-6} , respectively. GRPO uses rank 64, four epochs, learning rate 10^{-6} , $G = 5$, KL coefficient 0.01, clipping radius 0.2, and reward weights (0.10, 0.70, 0.05, 0.05, 0.05, 0.05) for Action Type, Action Argument, State Delta, Recovery, Progress/Loop, and Online Environment. The vision encoder is frozen throughout. Formal distributed training uses seven 8-GPU nodes (56 A100 80GB GPUs) with EFA; evaluation uses 15 8-GPU nodes and one tensor-parallel-size-1 vLLM instance per GPU. External checkpoints are evaluated with their released prompt, parser, coordinate convention, and native no-thinking/direct-action path. We report full sample sets, reject runs with missing samples or API errors, and retain raw predictions and per-task records.

4.1.2 BENCHMARKS AND METRICS

We evaluate four offline and two executable benchmarks. **AndroidControl** contains diverse demonstrations from 833 Android applications and provides high- and low-level instructions; we report action accuracy on 9,980 test steps per split (Li et al., 2024). **GUI-Odyssey** targets cross-application mobile navigation; its evaluation contains 83,885 steps over random, task, device, and app splits, and we report overall step success rate (SR), type match (TM), and grounding rate (GR) (Lu et al., 2025a). **ScreenSpot** contains 1,272 grounding queries across mobile, desktop, and web interfaces (Cheng et al., 2024), while **ScreenSpot-Pro** contains 1,581 queries from high-resolution professional applications (Li et al., 2025a); both use target-hit accuracy. **AndroidWorld** provides 116 reproducible task templates across 20 Android applications (Rawles et al., 2025); we execute three instances per template and report task success. **MobileWorld** stresses longer, cross-application workflows with reproducible backend verification (Kong et al., 2025); following our GUI-only protocol, we evaluate 117 tasks with one run each.

4.1.3 COMPARED MODELS

Qwen3.5-9B is a natively multimodal 9B foundation model with a hybrid linear/full-attention language backbone and serves as the unadapted base control (Qwen Team, 2026). We additionally compare seven released 7B–8B GUI agents. **Aguvis-7B-720P** is a cross-platform pure-vision agent trained in two stages for grounding and planning (Xu et al., 2025b). **GUI-R1-7B** applies GRPO with unified action-rule rewards to a compact curated set (Luo et al., 2025). **ScaleCUA-7B** scales cross-platform computer-use data over six operating systems (Liu et al., 2026c). **GUI-Libra-8B** combines action-aware supervision with KL-regularized partially verifiable RL (Yang et al., 2026). **OpenMobile-8B** synthesizes mobile tasks and trajectories through exploration and policy switching (Cheng et al., 2026). **MemGUI-8B-SFT** uses Context-as-Action to proactively fold long-horizon interaction history (Liu et al., 2026a). **ForgeQwen3-8B** is the Qwen3-VL checkpoint adapted by MobileForge using automatically generated target-app tasks and hierarchical trajectory/step feedback (Liu et al., 2026b).

4.2 OFFLINE BENCHMARK RESULTS

Under the action-only protocol, with parse failures treated as errors, Table 1 indicates that TRAVEC-9B improves joint action selection and grounding rather than optimizing either component in isolation. Native Qwen3.5-9B ranks strongly on GUI-Odyssey grounding but not on type matching or step success, showing that localization alone does not yield executable decisions. TRAVEC-9B preserves this grounding strength while improving the other two metrics, indicating tighter coupling among perception, action selection, and transition progress. ForgeQwen3-8B slightly leads in type matching, yet its weaker grounding and step success show that a correct action class remains insufficient without accurate arguments and trajectory-aware execution.

The advantage widens on AndroidControl Low, suggesting that TRAVEC-9B makes effective use of explicit action-level instructions. The smaller High margin indicates that higher-level task decomposition remains more challenging. Its ScreenSpot-Pro result extends this pattern to visually dense professional interfaces and shows transfer beyond mobile trajectories. TRAVEC-9B also retains the compact generation behavior of the base model while reducing latency and completion length. Specialized GUI baselines require slower and longer requests without matching its overall accuracy. This joint pattern supports moving verification and recovery reasoning into training while keeping the deployed policy action-only. Because Table 1 contains offline metrics, the evidence establishes per-action capability and efficiency rather than complete task execution.

4.3 ABLATING PROCESS-AUDIT AND SUPERVISION COMPONENTS

Table 2 isolates how process auditing and auxiliary supervision shape the rollout data used by the Action Policy.

Removing Action SFT weakens every evaluation regime, showing that audited GRPO depends on the action and interface prior learned during SFT. Other omissions reorder the benchmarks, indicating complementary components whose utility varies by task. Difficult grounding is most sensitive to structured contract fields, while static metrics expose little benefit from recovery supervision. Retaining more steps without Critic filtering offers no consistent gain, so the Critic improves rollout quality rather than data volume. Since the formal TRAVEC-9B reference uses a different schedule, it is not a matched causal baseline.

Table 2: Component ablations. “w/o” denotes *without*. Best and second-best results are bolded and underlined.

Model	AndroidControl		ScreenSpot	
	High ↑	Low ↑	SS ↑	Pro ↑
w/o Critic	58.048	82.097	92.452	63.061
w/o Contract	57.024	80.110	<u>92.610</u>	63.820
w/o Verifier	<u>59.573</u>	82.994	92.374	63.947
w/o Contract Fields	56.344	82.816	91.824	62.998
w/o Recovery	56.723	<u>86.553</u>	92.296	<u>64.643</u>
w/o Action SFT	49.453	74.170	89.937	56.610
TRAVEC-9B	64.309	93.337	92.689	67.805

4.4 ABLATING PROCESS-REWARD WEIGHT

Table 3 reveals a nonmonotonic balance between action fidelity and process shaping. Mixed objectives outperform the action only and all process settings on most metrics, showing that transition evidence complements rather than replaces direct action supervision. Changing process mass also shifts the preferred setting between instruction following and visual grounding, so more process supervision is not uniformly better. The formal TRAVEC-9B vector follows this pattern by keeping action correctness dominant and process rewards bounded. Its broad advantage is consistent with this balance, but the schedule mismatch and absence of a matched 20% run prevent a claim that the deployed vector is optimal.

Table 3: Reward-weight ablation. Weights follow the order Type, Argument, State Delta, Recovery, Progress/Loop, and Environment. The separated final row denotes the TRAVEC-9B configuration. Best and second-best results are bolded and underlined.

Reward weights	AndroidControl		ScreenSpot	
	High ↑	Low ↑	SS ↑	Pro ↑
(0.125, 0.875, 0, 0, 0, 0)	54.890	78.816	92.060	62.998
(0.0625, 0.4375, 0.125, 0.125, 0.125, 0.125)	<u>56.523</u>	82.425	92.453	<u>63.820</u>
(0.025, 0.175, 0.2, 0.2, 0.2, 0.2)	55.631	<u>82.745</u>	92.846	63.631
(0, 0, 0.25, 0.25, 0.25, 0.25)	55.952	81.152	92.060	63.188
(0.10, 0.70, 0.05, 0.05, 0.05, 0.05)	64.309	93.337	<u>92.689</u>	67.805

5 CONCLUSION

We presented TRAVEC, a framework that turns transition prediction, evidence-based verification, and failure recovery into scalable annotation and reinforcement-learning signals for mobile GUI agents. Its automated pipeline builds 550K verifiable contract, negative-recovery, preference, and calibration annotations; two auxiliary adapters and a trajectory Critic audit resettable self-generated experience; and six rewards align the final policy with action correctness, observed state change, recovery, progress, loop avoidance, and execution outcome. The resulting TRAVEC-9B removes all auxiliary modules at deployment and emits only executable actions. Across the completed experiments, it improves substantially over its Qwen3.5-9B base and Action SFT on AndroidWorld while remaining strong on offline action and grounding benchmarks; the component results also identify concrete trade-offs that the pending reward and efficiency evaluations must resolve. More broadly, TRAVEC offers a route to move expensive reasoning and verification away from the online critical path and into auditable policy learning.

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